Answer **one** question from this section.

8 (a) Fig. 8.1 shows a diagram of an electron gun.

Electrons are emitted from a hot cathode. The electric field between the cathode and the anode accelerates the electrons through an accelerating potential difference (p.d.).

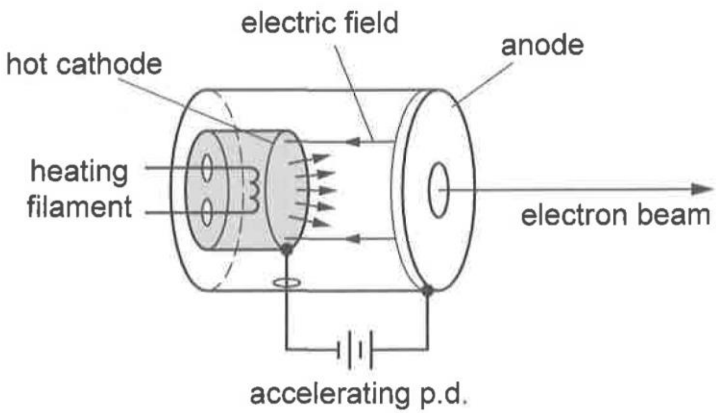


Fig. 8.1

(i) Explain what is meant by an electric field.

....

....[1]

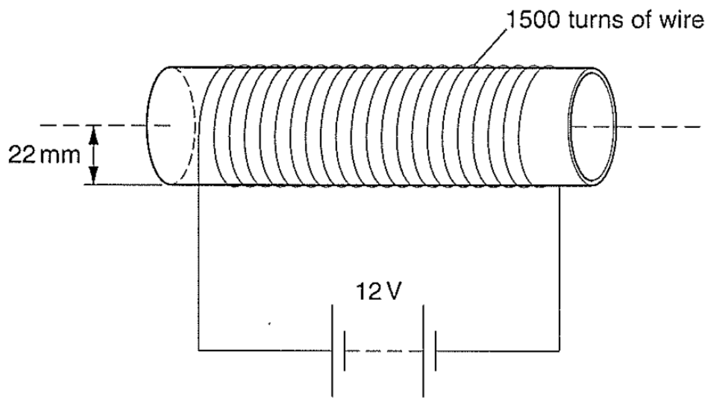
Solution:

Region of space where a charged body experiences an electric force | 1

(ii) Explain why the electron gun in Fig. 8.1 must be in a vacuum.

....

			……………………………………………………………………………………… ……………………………………………………………………………………… ……………………………………………………………………………………… [2]	
			Solution: Electrons have a small mass easily deflected/ weakly ionizing/ collide with air molecules and scatter In vacuum, they can travel in a straight line and reach the anode (if not they will be deflected from their path)	 1 1
		(iii)	The kinetic energy of the electrons increases by 5.68×10^{-16} J between leaving the cathode and reaching the anode. Calculate the accelerating p.d..	
			accelerating p.d. = …………………… V [2]	
			Solution: W = qV $5.68 \times 10^{-16} = 1.6 \times 10^{-19} \times V$ V = 3.55 kV	 1 1
		(iv)	Suggest why the electrons reaching the anode have a range of speeds.	
			……………………………………………………………………………………… ……………………………………………………………………………………… ……………………………………………………………………………………… ……………………………………………………………………………………… [2]	
			Solution: Thermionic emission of electrons from the filament is a random process and electrons are emitted with a range of KE from the filament.	 1

			When accelerated by the p.d., the increase in KE is the same, but since they had a range of KE when they were produced, the final KE when reaching the anode will have a range of values, hence a range of speeds.	1
	(b)	A uniform magnetic field is produced using a coil of 1500 turns of insulated wire, tightly wound on a non-magnetic tube to make a solenoid of mean radius 22 mm, as shown in Fig. 8.2. The wire itself has radius 0.86 mm and is made of a material of resistivity 1.7×10^{-8} Ω m. The coil is connected to a supply of e.m.f. 12 V and negligible internal resistance.  **Fig. 8.2** Calculate		
		(i)	the total length of the 1500 turns of wire in the coil, length = ... m [1]	
			Solution: total length of wire in the coil = $2\pi(22 \times 10^{-3})(1500) = 207.3$ m 1	
		(ii)	the total resistance of the coil, resistance = ... Ω [2]	
			Solution: $R = \frac{\rho l}{A} = \frac{1.7\times10^{-8}\times207.3}{\pi(0.86\times10^{-3})^2}$ 1 $= 1.52\ \Omega$ 1	

(iii) the current in the coil.

current = ... A [2]

Solution:

I = 12 / 1.52 1
= 7.91 A 1

(c) The magnetic flux density in the solenoid is measured using a current balance. The current balance is a U-shaped piece of stiff wire ABCDEF pivoted at BE, as shown in Fig. 8.3.

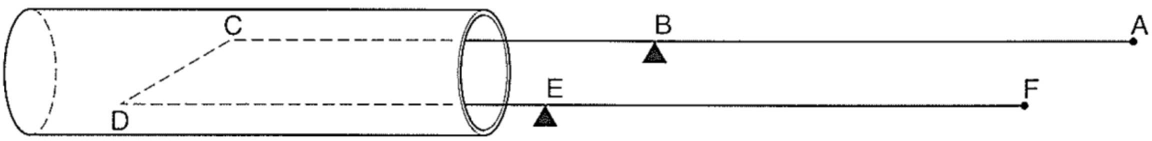


Fig. 8.3

When in use, there is a turning force on the stiff wire caused by a current in CD.

(i) Explain why the current in CD causes a turning effect.

....

....

.... [2]

Solution:

When there is a current in CD, it produces a magnetic field around the wire which interacts with the magnetic field of the solenoid. This results in a force perpendicular to both the current and the magnetic field of the solenoid. 1

This force x the perpendicular distance of wire CD from the pivot produces a moment and causes a turning effect. 1

(ii) Explain why currents in CB and DE do not contribute to the turning force.

....

.... [1]

Solution:

Currents in CB and DE are parallel to the magnetic field of the solenoid and hence no force. 1

(iii) CD has length 25 mm, CB and DE each have length 106 mm.

The stiff wire is first balanced when there is no current in it. A current of 4.9 A is then passed through CD and, in order to rebalance the stiff wire, a force of 5.7×10^{-4} N is applied at a distance of 77 mm from the pivot, as shown in Fig. 8.4. which is the side view of the balance.

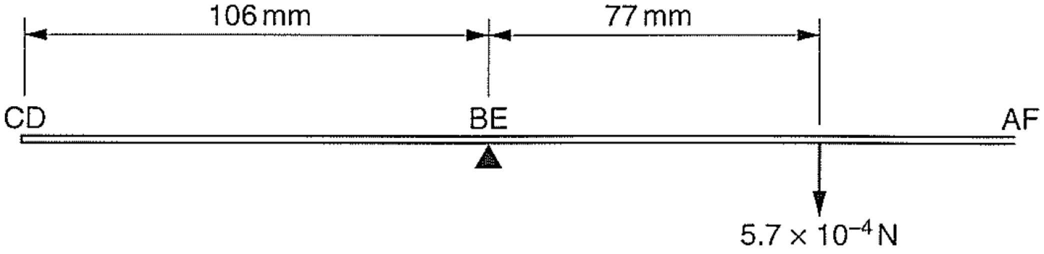


Fig. 8.4 (side view)

1. State the direction of the current in CD

direction = …………………… [1]

Solution: C to D 1

2. Calculate the magnetic flux density in the solenoid. Give the full name of the unit for magnetic flux density.

magnetic flux density = …………………………

full name of unit = …………………….. [4]

[Total: 20]

Solution:

By POM,
$F \times 0.106 = 5.7\times10^{-4} \times 0.077$ 1

And $F = BIL = B(4.9)(0.025)$ 1

$B = 3.38 \times 10^{-3}$ T 1

Tesla 1

9	**(a)**	State what is meant by *simple harmonic motion*.